Group A: By Interacting Object (Box, Sphere, Cylinder) — Motions: Kinematic (Horizontal, Vertical, Circle, Hold) & Dynamic (Drop, Slide, Throw, Tumble)

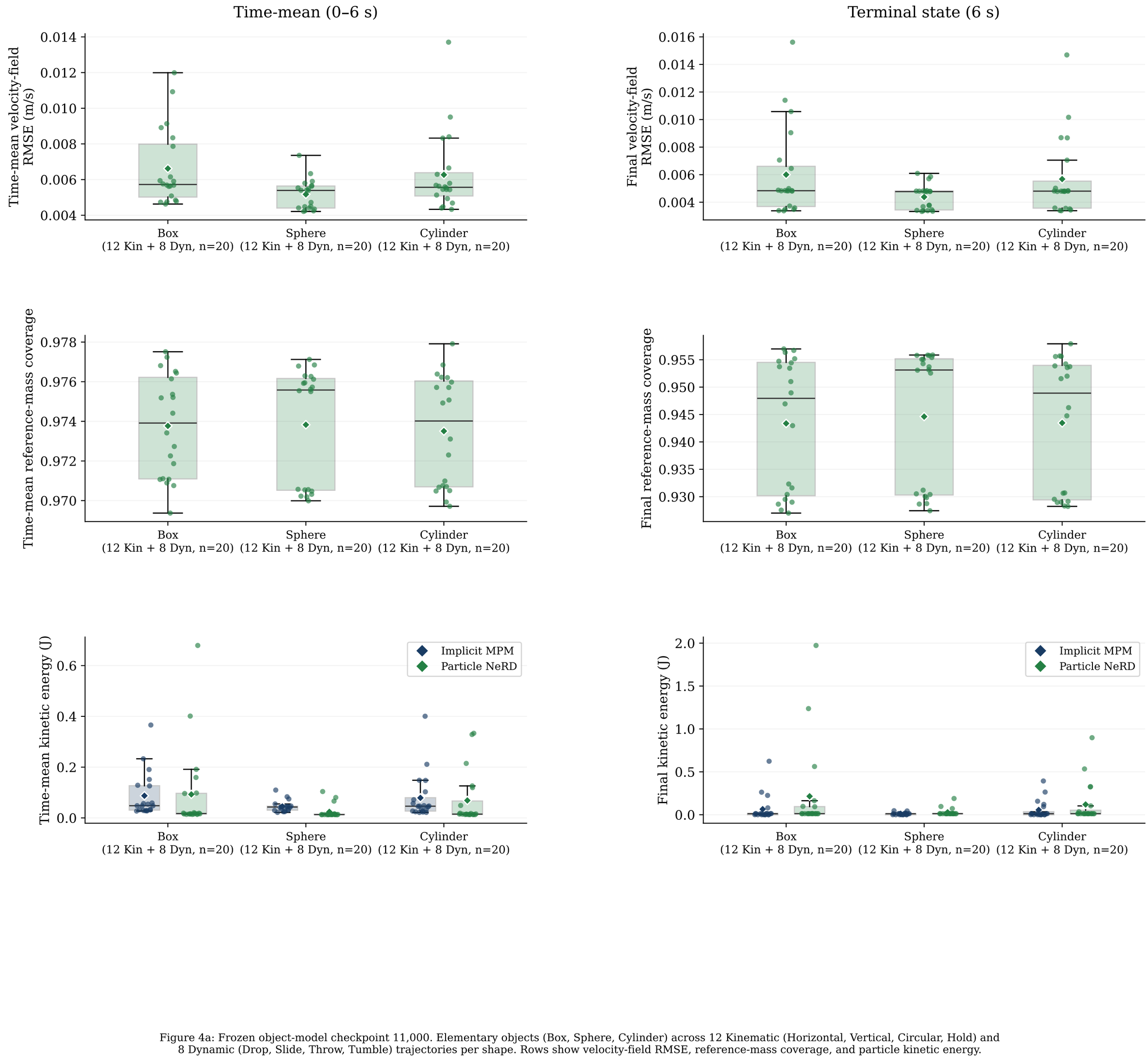

Figure 4a: Frozen object-model checkpoint 11,000. Elementary objects (Box, Sphere, Cylinder) across 12 Kinematic (Horizontal, Vertical, Circular, Hold) and 8 Dynamic (Drop, Slide, Throw, Tumble) trajectories per shape. Rows show velocity-field RMSE, reference-mass coverage, and particle kinetic energy.